Progress Report: Sprint 13

CS.028 Optimizing Research Software Codes - 10 May 2026

[Kathryn Butler](), [Joseph Schaab](), [Michael McAllister]()

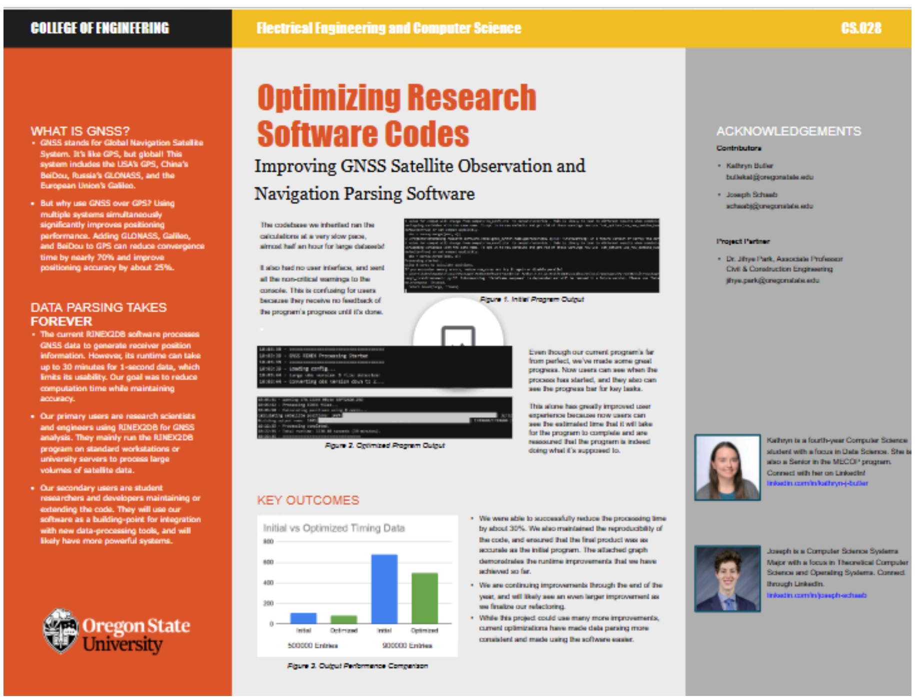
Summary

- Finished logical improvements to calculate_positions(). Function now uses multi-dimensional memory access.
- Implemented testing using the newly provided small data files.
- Completed our poster's final draft. Awaiting approval for printing.

Sprint Work Log

Poster

Status: Complete
Authors: Kathryn Butler, Joseph Schaab
Reviewers: N/A
Source Link: [Commit 7c03c12]()
Supporting Artifact:



Notes: Confirmed that our project meets the requirements, and conducted extensive testing to make sure our timing results were accurate.

Config Reading From Directory

Status: In Progress (30% complete)
Authors: Kathryn Butler

Reviewers: N/A
Source Link: [Issue 16]
Supporting Artifact:

```yaml
observation:
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2620.25O
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2622.25O
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2624.25O
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2626.25O
navigation:
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2621.25P
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2623.25P
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2625.25P
  - C:\Projects\optimizing-research-software-codes\Datasets\COVL_1sec\SEPT2627.25P
```

Notes: We have to input every single observation and navigation file. This is not user-friendly. Instead, the user should just place the files in an observation/navigation directory, and the program reads from them and outputs them into a file. This can be done with a separate directory for observation and navigation or just all in one.

Finalized Calculation Logic

Status: Complete
Author: Joseph Schaab
Reviewers: N/A
Source Link: calculate_positions() - Final Logic
Supporting Artifact: Testing confirmed vectorized call + itertuples() is on average ten times faster than the original apply(lambda x: ..., axis=1) method.
Notes: Tested logical changes to calculate_positions() using the three different sizes of testing data. After isolating the runtime used by the modules of interest, the difference in performance was observed as follows:

- Small size data ran 7.3 times faster.
- Medium size data ran 8.6 times faster.
- Large size data ran 10.4 times faster.

Risk and Quality Tracking

Bug Count: 11 (Link to issue query)
Top Bugs:

- Position Variable Mutation
  - **Severity:** High
  - **Owner:** Kathryn Butler
  - **Current Status:** Identified, waiting on meeting
  - **Actions:** Meet with Dr. Park to ask about the potential calculation issue.
- Treating Keys as Positions and Using Argsort

    - **Severity:** High
    - **Owner:** Kathryn Butler
    - **Current Status:** Resolved
    - **Actions:** No further action needed. (Series.getitem treating keys as positions is deprecated. Look up how to solve it.)
- Incorrect conversion from float to datetime
    - **Severity:** Medium
    - **Owner:** Kathryn Butler
    - **Current Status:** Resolved
    - **Actions:** No further action needed. (Research into the line causing the problem. Determine the input causing the problem, and correct the datetime variable.)
- Log.txt hardcoded
    - **Severity:** Medium
    - **Owner:** Kathryn Butler
    - **Current Status:** Resolved
    - **Actions:** No further action needed. (Log.txt seems to be announcing a preset number of cores to calculate. Figure out why it is set to only be 8 cores. Find out if you can manually adjust the core count to be something other than just 8, then add the text to the function dynamically.)
- Inconsistent timing data
    - **Severity:** High
    - **Owner:** Kathryn Butler
    - **Current Status:** Ongoing
    - **Actions:** Create a smaller test data pool, run the tests multiple times to determine if the size of the file increases the likelihood of failed parsing attempts.
- Xarray.merge conflict
    - **Severity:** Medium
    - **Owner:** Kathryn Butler
    - **Current Status:** Ongoing
    - **Actions:** Set compat explicitly to something other than ‘override’, then test output for clarity. (Previous goal: Research what “compat” does, and either set it explicitly or use the new combine_kwarg_defaults.)

Next Sprint Plan

- Complete the streaming refactor. Finish the chunked calculations and simplify the progress bar. Source of truth: Commit 9bdb23f
- Resolve position variable mutation. Meet with Dr. Park to review the code to check if there is an error. Source of truth: Issue 15

- Stretch goal: Allow users to drop RINEX files into a folder and have the program detect it from there instead of manually updating the config.yaml file. Source of truth: Issue 16

Team Process Reflection

In this sprint, we were able to implement more refined testing due to new data sets that were recently provided to us by Dr. Park. These new data files are small in size so they can be run on our systems much quicker than our previous testing data. This has greatly benefited our development and debugging by allowing us to test the program much faster. We are expecting this to positively support our future work since we will be able to make more detailed improvements with the time saved from testing the large data sets. We will know this addition has proven to be successful if we can demonstrate runtime improvements to all three testing data sizes: small(new), medium, and large.

We were also able to benefit from helpful information from the program's original developer. Our previous uncertainties concerning rnx2db.py have now been cleared up. This has helped us understand the program more completely and in more depth. This will support our documentation to help the future developers of this program. We plan to continue working on this project through the end of the term to deliver a more polished final product.